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Straßenbeleuchtung - Teil 3: Berechnung der Gütemerkmale

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Foreword

This document (EN 13201-3:2003) has been prepared by Technical Committee CEN/TC 169 “Light and lighting”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by May 2004, and conflicting national standards shall be withdrawn at the latest by May 2004.

This European Standard was worked out by the Joint Working Group of CEN/TC 169 "Light and lighting" and CEN/TC 226 "Road Equipment", the secretariat of which is held by AFNOR.

This document includes a Bibliography.

This standard, EN 13201 *Road lighting*, consists of three parts. This document is:

Part 3: *Calculation of performance*

The other parts of EN 13201 are:

Part 2: *Performance requirements*

Part 4: *Methods of measuring lighting performance*

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Slovakia, Spain, Sweden, Switzerland and the United Kingdom.

Introduction

The calculation methods described in this Part of this European Standard enable road lighting quality characteristics to be calculated by agreed procedures so that results obtained from different sources will have a uniform basis.

1 Scope

This European Standard defines and describes the conventions and mathematical procedures to be adopted in calculating the photometric performance of road lighting installations designed in accordance with EN 13201-2.

2 Normative references

This European Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text, and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies (including amendments).

prEN 13032-1, *Light and lighting — Measurement and presentation of photometric data of lamps and luminaires — Part 1: Measurement and file format.*

EN 13201-2, *Road lighting — Part 2: Performance requirements.*

3 Terms, definitions, symbols and abbreviations

3.1 Terms and definitions

For the purposes of this European Standard, the following terms and definitions apply.

3.1.1
vertical photometric angle (of a light path) (γ)
angle between the light path and the first photometric axis of the luminaire

NOTE 1 Unit °(degrees).

NOTE 2 See Figure 1.

3.1.2
azimuth (of a light path) (*C*)
angle between the vertical half-plane passing through the light path and the zero reference half-plane through the first photometric axis of a luminaire, when the luminaire is at its tilt during measurement

NOTE 1 Unit ° (degrees).

NOTE 2 See Figure 1.

3.1.3

angle of incidence (of a light path at a point on a surface) (ε)

angle between the light path and the normal to the surface

NOTE 1 Unit ° (degrees).

NOTE 2 See Figure 4, Figure 13 and Figure 14.

3.1.4

angle of deviation (with respect to luminance coefficient) (β)

supplementary angle between the vertical plane through the luminaire and point of observation and the vertical plane through the observer and the point of observation

NOTE 1 Unit °(degrees).

NOTE 2 See Figure 4.

3.1.5

luminance coefficient (at a surface element, in a given direction, under specified conditions of illumination) (*q*)

quotient of the luminance of the surface element in the given direction by the illuminance on the medium

NOTE 1 Unit sr^{-1}

NOTE 2 $$q = \frac{L}{E} \qquad (1)$$

where:

q is the luminance coefficient, in reciprocal steradians

L is the luminance, in candelas per square metre

E is the illuminance, in lux

3.1.6

reduced luminance coefficient (for a point on a surface) (*r*)

luminance coefficient multiplied by the cube of the cosine of the angle of incidence of the light on the point

NOTE 1 Unit sr^{-1}

NOTE 2 This can be expressed by the equation: $r = q \cos^3 \varepsilon$ (2)

where:

q is the luminance coefficient, in reciprocal steradians

ε is the angle of incidence, in degrees

NOTE 3 The angle of observation, α in Figure 4, affects the value of r. By convention this angle is fixed at 1° for road lighting calculations. r is reasonably constant for values of α between 0,5° and 1,5°, the angles over which luminance calculations for the road surface are generally required.

3.1.7

tilt during measurement (of a luminaire) (θ_m)

angle between a defined datum axis on the luminaire and the horizontal when the luminaire is mounted for photometric measurement

NOTE 1 Unit °(degrees).

NOTE 2 See Figure 8.

NOTE 3 The defined datum axis can be any feature of the luminaire, but generally for a side-mounted luminaire it lies in the mouth of the luminaire canopy, in line with the spigot axis. Another commonly used feature is the spigot entry axis.

3.1.8

tilt in application (of a luminaire) (θ_f)

angle between a defined datum axis on the luminaire and the horizontal when the luminaire is mounted for field use

NOTE 1 Unit °(degrees).

NOTE 2 See Figure 1 and Figure 8.

NOTE 3 The defined datum axis can be any feature of the luminaire but generally for a side-mounted luminaire it lies in the mouth of the luminaire canopy, in line with the spigot axis. Another commonly used feature is the spigot entry axis.

3.1.9

orientation (of a luminaire) (ν)

angle a chosen reference direction makes with the C = 0°, γ = 90° measurement direction of the luminaire when the first photometric axis of the luminaire is vertical

NOTE 1 Unit °(degrees).

NOTE 2 When the road is straight the reference direction is longitudinal.

NOTE 3 See Figure 7, which illustrates the sign conventions.

3.1.10

rotation (of a luminaire) (ψ)

angle the first photometic axis of the luminaire makes with the nadir of the luminaire, when the tilt during measurement is zero

NOTE 1 Unit °(degrees).

NOTE 2 See Figure 7, which illustrates the sign conventions.

3.1.11

first photometric axis (of a luminaire when measured in the (C, γ) coordinate system)

vertical axis through the photometric centre of a luminaire when it is at its tilt during measurement

NOTE 1 The poles of the (C,γ) coordinate system lie in this axis. See Figure 1.

NOTE 2 This axis is tilted when the luminaire is tilted from its tilt during measurement.

3.1.12

longitudinal direction

direction parallel to the axis of the road

3.1.13

transverse direction

direction at right angles to the axis of the road

NOTE On a curved road the transverse direction is that of the radius of curvature at the point of interest on the road.

3.1.14
installation azimuth (with respect to a given point on the road surface and a given luminaire at its tilt during measurement) (φ)
angle a chosen reference direction (which is longitudinal for a straight road) makes with the vertical plane through the given point and the first photometric axis of the luminaire, when the luminaire is at its tilt during measurement

NOTE 1 Unit °(degrees).

NOTE 2 See Figure 4.

3.2 List of symbols and abbreviations

The symbols and abbreviations used in this standard are listed in Table 1.

Table 1 — Symbols and abbreviations

Quantity		
Symbol	Name or description	Unit
C	Photometric azimuth (Figure 1)	°(degrees)
D	Spacing between calculation points in the longitudinal direction	m
d	Spacing between calculation points in the transverse direction	m
E	Illuminance	lx
H	Mounting height of a luminaire	m
j,m	Integers indicating the row or column of a table	-
L	Luminance	cd/rn^2
I	Luminous intensity per kilolumen	cd/klm
L_p	Total luminance at a point P	cd/m^2
MF	Product of the lamp flux maintenance factor and the luminaire maintenance factor	-
N	Number of points in the longitudinal direction	-
n	Number of luminaires considered in the calculation	-
q	Luminance coefficient	sr^{-1}
Q_0	Average luminance coefficient	sr^{-1}
r	Reduced luminance coefficient	sr^{-1}
S	Spacing between luminaires	m
TI	Threshold increment	%
L_v	Equivalent veiling luminance	cd/m^2
W_L	Width of driving lane	m
W_r	Width of relevant area	m
W_S	Width of strip	m
x	Abscissa in *(x,y)* coordinate system (Figure 6)	m
y	Ordinate in *(x,y)* coordinate system (Figure 6)	m
Φ	Luminous flux of lamp or lamps in a luminaire	klm
α	Angle between the incident light path and the normal to the flat surface of the semicylinder used for measuring semicylindrical illuminance (Figure 13), or the designated vertical plane used for vertical illuminance (Figure 14)	° (degrees)
β	Angle of deviation (Figure 4)	° (degrees)
γ	Vertical photometric angle (Figure 1)	° (degrees)
δ	Tilt for calculation (Figure 8)	° (degrees)
ε	Angle of incidence (Figure 4)	° (degrees)
θ_1	Tilt in application (Figure 8)	° (degrees)
θ_m	Tilt during measurement (Figure 8)	° (degrees)
v	Orientation of luminaire (Figure 7)	° (degrees)
σ	Angle of observation (Figure 4)	° (degrees)
φ	Installation azimuth (Figure 4)	° (degrees)
Ψ	Rotation of a luminaire (Figure 1)	° (degrees)

4 Mathematical conventions

The basic conventions made in the mathematical procedures described in this standard are:

— the luminaire is regarded as a point source of light;

— light reflected from the surrounds and interreflected light is disregarded;

— obstruction to the light from luminaires by trees and other objects is disregarded;

— the atmospheric absorption is zero;

— the road surface is flat and level and has uniform reflecting properties over the area considered.

5 Photometric data

5.1 General

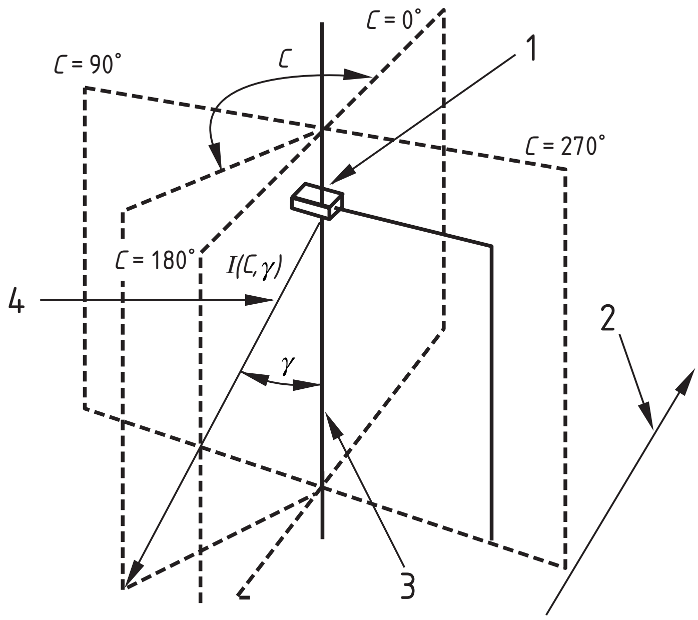
Photometric data for the light distribution of the luminaires used in the lighting installation are needed for calculating the lighting quality characteristics in this standard. These data are in the form of an intensity table ($\mathcal{I}$-table) which gives the distribution of luminous intensity emitted by the luminaire in all relevant directions. When luminance calculations are to be made, photometric data for the light reflecting properties of the road surface are required in the form of an *r*-table.

Interpolation will be needed in using both these tables to enable values to be estimated for directions between the tabulated angles.

5.2 The $\mathcal{I}$-table

For calculations made to this standard, an intensity table *($\mathcal{I}$-table)* prepared in accordance with prEN 13032-1 is required. The coordinate system used for road lighting luminaires is the *(C, γ),* shown in Figure 1, although the (B, β) coordinate system may be used for floodlights. In the figure, the luminaire is shown at its tilt during measurement.

Luminous intensity shall be expressed in candelas per kilolumen (cd/klm) from all the light sources in the luminaire.



Key

1 Luminaire at tilt during measurement

2 Longitudinal direction

3 First photometric axis

4 Direction of luminous intensity

Figure 1 — Orientation of *C,γ* coordinate system in relation to longitudinal direction of carriageway

Maximum angular intervals stipulated in this standard have been selected to give acceptable levels of interpolation accuracy when the recommended interpolation procedures are used.

In the (*C,γ*) system of coordinates, luminous intensities shall be provided at the angular intervals stated below.

For all luminaires the angular intervals in vertical planes (*γ*) shall at most be 2,5° from 0° to 180°. In azimuth the intervals shall be varied according to the symmetry of the light distribution from the luminaire as follows:

a) luminaires with no symmetry about the *C* = 0° plane: the intervals shall at most be 5°, starting at 0°, when the luminaire is at its tilt during measurement, and ending at 355°;

b) luminaires with nominal symmetry about the *C* = 270° - 90° plane: the intervals shall at most be 5°, starting at 270°, when the luminaire is at its tilt during measurement, and ending at 90°;

c) luminaires with nominal symmetry about the *C* = 270° - 90° and *C* = 0° - 180° planes: the intervals shall at most be 5°, starting at 0°, when the luminaire is at its tilt during measurement, and ending at 90°;

d) luminaires with nominally the same light distribution in all *C*-planes: only one representative set of measurements in a vertical (*C*-plane) is needed.

NOTE The angular spacings recommended in CIE Publication 140 for *I*-tables are wider than those recommended above, and may not give results which are of a satisfactory accuracy for illuminance calculations.

5.3 Interpolation in the *I-table*

5.3.1 General

Where the intensity is required in a direction which does not lie in one of the directions in which measurements are recorded, either linear or quadratic, interpolation will be necessary to estimate the intensity in the desired direction. Linear interpolation is the simpler procedure and may be used where the angular intervals are in accordance with those stipulated in 5.2. If the angular intervals are greater then it will be necessary to use quadratic interpolation.

5.3.2 Linear interpolation

To estimate the luminous intensity $I(C,\gamma)$ in the direction (C,γ), it is necessary to interpolate between four values of luminous intensity lying closest to the direction, see Figure 2.

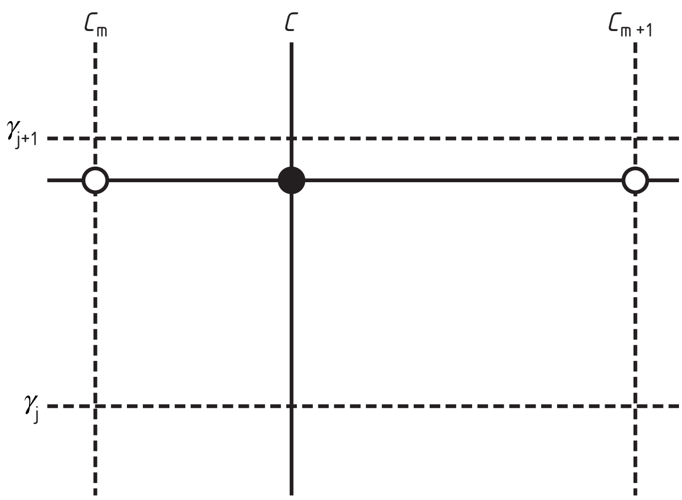


Figure 2 — Angles required for linear interpolation of luminous intensity

For this purpose, the following equations or mathematically equivalent equations shall be used:

$$K_1 = \frac{C_m - C}{C_m - C_{m+1}} \tag{3}$$

$$K_2 = \frac{\gamma_j - \gamma}{\gamma_j - \gamma_{j+1}} \tag{4}$$

where:

K_1 and K_2	are constants determined from the equations
C	is the azimuth, measured about the first photometric axis
γ	is the vertical angle measured from the first photometric axis
j, j+1, m, m+1	are integers indicating the number of the column or row in the I-table

$$I(C,\gamma_j) = I(C_m,\gamma_j) - K_1 \times [I(C_m,\gamma_j) - I(C_{m+1},\gamma_j)] \quad (5)$$

$$I(C,\gamma_{j+1}) = I(C_m,\gamma_{j+1}) - K_1 \times [I(C_m,\gamma_{j+1}) - I(C_{m+1},\gamma_{j+1})] \quad (6)$$

$$I(C,\gamma) = I(C,\gamma_j) - K_2 \times [I(C,\gamma_j) - I(C,\gamma_{j+1})] \quad (7)$$

where:

$I(C_m,\gamma_j)$	indicates the intensity in column number m and row number j of the I-table, and so on for the other similar symbols.

In these equations interpolation is first carried out in the γ cones, and then in the C-planes. If desired this procedure can be reversed (that is, the interpolation is first carried out in the C-planes followed by the γ cones) and the same result obtained.

5.3.3 Quadratic interpolation

Quadratic interpolation requires three values in the I-table for each interpolated value. Figure 3 indicates the procedure. If a value of I is required at (C, γ), interpolation is first carried out down three adjacent columns of the I-table enclosing the point. This enables three values of I to be found at γ. Interpolation is then carried out across the table to find the required value at (C, γ). If preferred, this procedure may be reversed; that is, interpolation can be carried out across and then down the I-table without affecting the result.

To reduce interpolation inaccuracies as far as possible the following two rules should be followed in selecting the values for insertion in the interpolation equations:

1) The two tabular angles adjacent to the angle for interpolation are selected for insertion in the interpolation equations and the average calculated.

2) If the angle for interpolation is smaller than this average then the third tabular angle is the next lower tabular angle (as shown for C in Figure 3); if the angle for interpolation is greater than this average then the third tabular angle is the next higher tabular angle (as shown for γ in Figure 3).

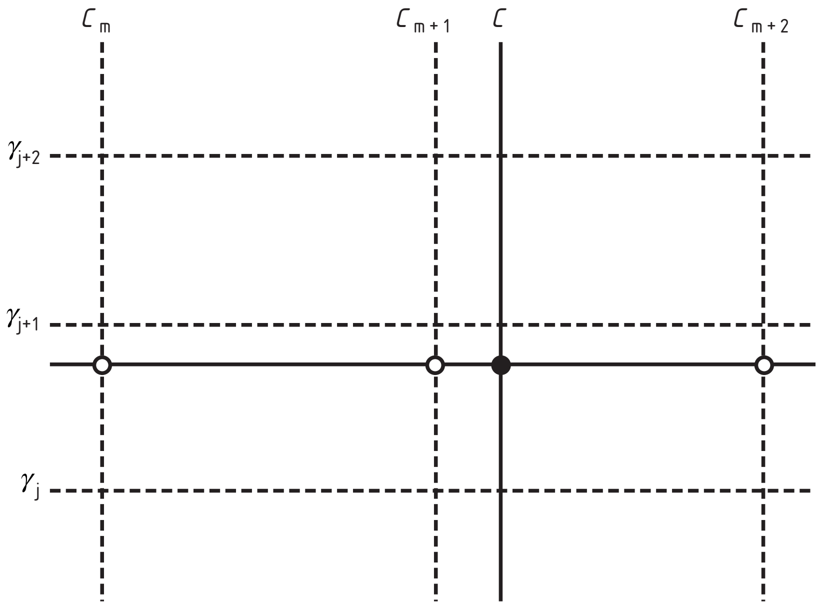


Figure 3 — Values required for quadratic interpolation

When interpolation is carried out in the region of C = 0°, or y = 0° or 180°, see 5.3.4.

The formula for quadratic interpolation is

$$y(x) = y_1\left(\frac{(x-x_2)(x-x_3)}{(x_1-x_2)(x_1-x_3)}\right) + y_2\left(\frac{(x-x_1)(x-x_3)}{(x_2-x_1)(x_2-x_3)}\right) + y_3\left(\frac{(x-x_1)(x-x_2)}{(x_3-x_1)(x_3-x_2)}\right) \quad (8)$$

where it will be noticed that there is cyclic permutation of the suffices.

This interpolation can be applied to either C or γ. When it is first applied to C, this parameter is substituted for x in the above equation:

$x = C$

$x_1 = C_m$

$x_2 = C_{m+1}$

$x_3 = C_{m+2}$

where:

C	is the angle at which I is to be found by interpolation
m, m+1, m+2	are integers indicating the number of the columns in the I-table
C_m, C_{m+1} and C_{m+2}	are values of C for the corresponding column numbers

From this substitution three constants can be defined, which can be conveniently evaluated by a subroutine program:

$$K_1 = \frac{(C - C_{m+1})(C - C_{m+2})}{(C_m - C_{m+1})(C_m - C_{m+2})} \quad (9)$$

$$K_2 = \frac{(C - C_m)(C - C_{m+2})}{(C_{m+1} - C_m)(C_{m+1} - C_{m+2})} \quad (10)$$

$$K_3 = \frac{(C - C_m)(C - C_{m+1})}{(C_{m+2} - C_m)(C_{m+2} - C_{m+1})} \quad (11)$$

From these three equations it follows that $K_1 + K_2 + K_3 = 1$. A set of three equations can then be written allowing evaluation in a calculation loop in a computer program with the variation of j:

$$I(C, \gamma_j) = K_1 I(C_m, \gamma_j) + K_2 I(C_{m+1}, \gamma_j) + K_3 I(C_{m+2}, \gamma_j) \quad (12)$$

$$I(C, \gamma_{j+1}) = K_1 I(C_m, \gamma_{j+1}) + K_2 I(C_{m+1}, \gamma_{j+1}) + K_3 I(C_{m+2}, \gamma_{j+1}) \quad (13)$$

$$I(C, \gamma_{j+2}) = K_1 I(C_m, \gamma_{j+2}) + K_2 I(C_{m+1}, \gamma_{j+2}) + K_3 I(C_{m+2}, \gamma_{j+2}) \quad (14)$$

For interpolation of the γ angles further application of Equation (3) gives three new constants:

$$k_1 = \frac{(\gamma - \gamma_{j+1})(\gamma - \gamma_{j+2})}{(\gamma_j - \gamma_{j+1})(\gamma_j - \gamma_{j+2})} \quad (15)$$

$$k_2 = \frac{(\gamma - \gamma_j)(\gamma - \gamma_{j+2})}{(\gamma_{j+1} - \gamma_j)(\gamma_{j+1} - \gamma_{j+2})} \quad (16)$$

$$k_3 = \frac{(\gamma - \gamma_j)(\gamma - \gamma_{j+1})}{(\gamma_{j+2} - \gamma_j)(\gamma_{j+2} - \gamma_{j+1})} \quad (17)$$

From these three equations it follows that $k_1 + k_2 + k_3 = 1$ and:

$$I(C, \gamma) = k_1 I(C, \gamma_j) + k_2 I(C, \gamma_{j+1}) + k_3 I(C, \gamma_{j+2})$$

which gives the required value of luminous intensity.

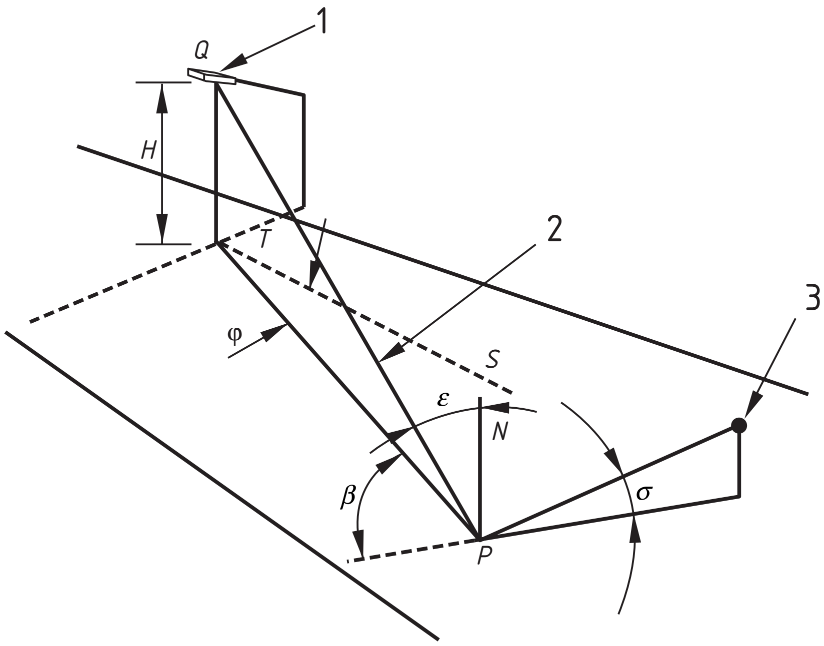
The order of the interpolation procedure, first for γ and then for C, may be reversed without altering the result.

5.3.4 Quadratic interpolation in the region of $C = 0°$, or $\gamma = 0°$ or 180°

For quadratic interpolation in these regions it may be necessary to take the third value of luminous intensity from the $C = 90°$ through to $C = 180°$ to 270° hemisphere of the luminous intensity distribution, which may be regarded as a mirror image of the $C = 270°$ through to $C = 0°$ to 90° hemisphere of the luminous intensity distribution.

5.4 The r-table

Road surface reflection data shall be expressed in terms of the reduced luminance coefficient multiplied by 10 000, at the angular intervals and in the directions given in Table 2 for the angles β and ε indicated in Figure 4.



Key

H Mounting height of the luminaire

PN N normal at P to the road surface

Q Photometric centre of the luminaire

QT First photometric axis of the luminaire

ST Longitudinal direction

β Supplementary angle

ε Angle of incidence

σ Angle of observation

φ Installation azimuth

1) Luminaire

2) Light path

3) Observer

Figure 4 — Angular relationships for luminaire at tilt during measurement, observer, and point of observation

Table 2 — Angular intervals and directions to be used in collecting road surface reflection data

tan ε	β in degrees																			
	0	2	5	10	15	20	25	30	35	40	45	60	75	90	105	120	135	150	165	180
0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0,25	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0,5	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0,75	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
1,25	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
1,5	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
1,75	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2,5	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
3	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
3,5	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
4	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
4,5	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
5	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
5,5	X	X	X	X	X	X	X	X	X											
6	X	X	X	X	X	X	X	X	X											
6,5	X	X	X	X	X	X	X	X												
7	X	X	X	X	X	X	X	X												
7,5	X	X	X	X	X	X	X													
8	X	X	X	X	X	X	X													
8,5	X	X	X	X	X	X	X													
9	X	X	X	X	X	X														
9,5	X	X	X	X	X	X														
10	X	X	X	X	X	X														
10,5	X	X	X	X	X	X														
11	X	X	X	X	X	X														
11,5	X	X	X	X	X															
12	X	X	X	X	X															

5.5 Interpolation in the *r*-table

When a value of *r* is required for values of tan ε and β lying between those given in the *r*-table it is necessary to use quadratic interpolation. This requires three values in the *r*-table for each interpolated value. Figure 5 indicates the procedure. If a value of *r* is required at (tan ε, β) interpolation is first carried out down three adjacent columns of the *r*-table enclosing the point. This enables three values of *r* to be found at tan ε. Interpolation is then carried out across the table to find the required value at (tan ε, β).

To reduce interpolation inaccuracies as far as possible the following rule shall be followed in selecting the values for insertion in the interpolation equations.

The two tabular values adjacent to the value for interpolation shall be selected. The third tabular value shall be the next greatest, shown in Figure 5. Linear interpolation shall be used at the boundaries of the table.

The ensuing mathematical procedure is similar to that described for the I-table (5.3.3).

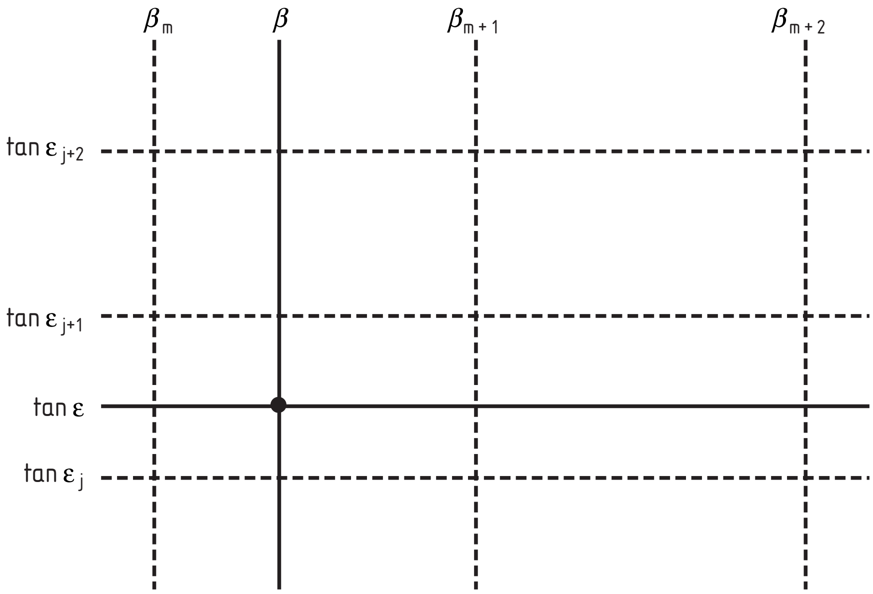


Figure 5 — Values required for interpolation procedure in the r-table

6 Calculation of $I(C,\gamma)$

6.1 General

To determine the luminous intensity from a luminaire to a point it is necessary to find the vertical photometric angle (γ) and photometric azimuth (C) of the light path to the point. To do this, account has to be taken of the tilt in application in relation to the tilt during measurement, the orientation, and rotation of the luminaire. For this purpose it is necessary to establish mathematical sign conventions for measuring distances on the road and for rotations about axes. The system used is a right-handed Cartesian coordinate system. The corrections for turning movements do not allow for any change in the luminous flux of the light source due to turning movements.

6.2 Mathematical conventions for distances measured on the road

A (x,y) rectangular coordinate system is used (Figure 6). The abscissa is aligned with the reference direction, which, for a straight road, lies in the longitudinal direction. Then:

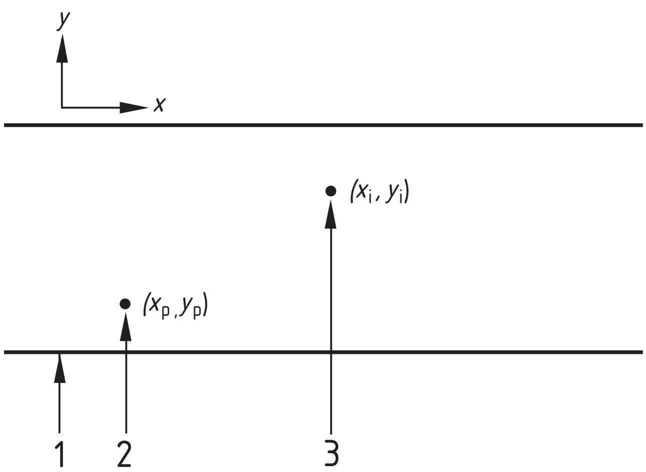
$$x = x_p - x_l \qquad (18)$$

$$y = y_p - y_l \qquad (19)$$

where:

(x_p, y_p) are the coordinates of the calculation point

(x_l, y_l) are the coordinates of the luminaire



Key

1 Edge of carriageway

2 Calculation point

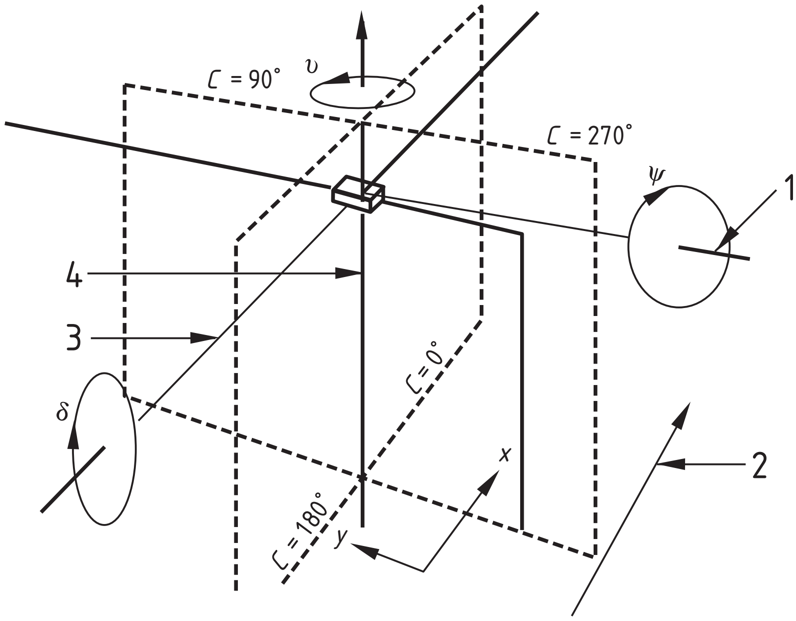
3 Luminaire

Figure 6 — (*x*,*y*) coordinate system for locating luminaire in plan

6.3 Mathematical conventions for rotations

Figure 7 shows the axes of rotation in relation to the (x,y) coordinate system, and the sense of the rotations.

Axis I is fixed in space, axis II and axis III can be turned about axis I.



Key

1 Axis III

2 Longitudinal direction

3 Axis II

4 First photometric axis I

Figure 7 — Axes of rotation in relation to (*x*,*y*) coordinate system

Figure 8 shows the relation of tilt for calculation to tilt during measurement and tilt in application. From this it is evident that:

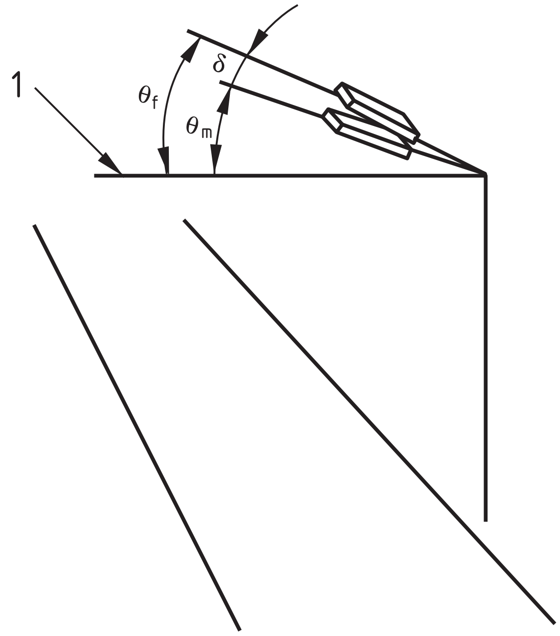
$$\delta = \theta_f - \theta_m \qquad (20)$$

where:

δ is the tilt in degrees for calculation

θ_f is the tilt in degrees in application

θ_m is the tilt in degrees during measurement



Key

δ	Tilt for calculation
θ_{f}	Tilt in application
θ_{m}	Tilt during measurement

1 Horizontal

Figure 8 — Tilt during measurement, tilt in application, tilt for calculation

6.4 Calculation of *C* and γ

These can be determined in four stages:

1) Substitution of v, ψ, δ, x and y in the equations:

$$x' = x(\cos v \ \cos\psi - \sin v \ \sin\delta \ \sin\psi) + y(\sin v \ \cos\psi + \cos v \ \sin\delta \ \sin\psi) + H \ \cos\delta \ \sin\psi \tag{21}$$

$$y' = -x \ \sin v \ \cos\delta + y \ \cos v \ \cos\delta - H \ \sin\delta \tag{22}$$

$$H' = -x(\sin v \ \sin\delta \ \cos\psi + \cos v \ \sin\psi) - y(\sin v \ \sin\psi - \cos v \ \sin\delta \ \cos\psi) + H \ \cos\delta \ \sin\psi \tag{23}$$

where:

x and y	are the longitudinal and transverse distances between the calculation point and the nadir of the luminaire in Figure 6
H	is the height of the luminaire above the calculation point

x', y', H' are the distances used for calculating C and γ; and may be regarded simply as intermediate variables

v, δ, and Ψ are the orientation, tilt for calculation, and rotation

2) Evaluation of installation azimuth φ.

Evaluation of arctan $\frac{Y}{X}$ will give:

$$-90° \leq \arctan \frac{Y}{X} \leq 90° \quad (24)$$

The angular quadrant in which this lies is determined by:

For $x > 0$, $y > 0$ $\quad \varphi = \arctan\frac{y}{x} \quad or = 0° \leq \varphi \leq 90°$ (25)

For $x < 0$, $y > 0$ $\quad \varphi = 180° + \arctan\frac{y}{x} \quad or = 90° \leq \varphi \leq 180°$ (26)

For $x < 0$, $y < 0$ $\quad \varphi = 180° + \arctan\frac{y}{x} \quad or = 180° \leq \varphi \leq 270°$ (27)

For $x > 0$, $y < 0$ $\quad \varphi = 360° + \arctan\frac{y}{x} \quad or = 270° \leq \varphi \leq 360°$ (28)

3) Calculation of C

$$C = \varphi - v \quad (29)$$

where:

φ is the installation azimuth in degrees

v is the orientation in degrees (Figure 7), obtained from the equations in 6.4, x' and y' being used in place of x and y respectively.

4) Calculation of γ

$$\gamma = \tan^{-1}\frac{\sqrt{(x')^2 + (y')^2}}{H'} \quad (30)$$

7 Calculation of photometric quantities

7.1 Luminance

7.1.1 Luminance at a point

The luminance at a point shall be determined by applying the following formula or a mathematically equivalent formula:

$$L = \frac{I \times r \times \Phi \times MF \times 10^{-4}}{H^2} \tag{31}$$

where:

L is the maintained luminance, in candelas per square metre

I is the luminous intensity in the direction (C, γ), indicated in Figure 1 and Figure 4, in candelas per kilolumen

r is the reduced luminance coefficient for a light path incident with angular coordinates (ε,β), in reciprocal steradians

Φ is the initial luminous flux of the sources in each luminaire, in kilolumens

MF is the product of the lamp flux maintenance factor and the luminaire maintenance factor

H is the mounting height of the luminaires above the surface of the road, in metres

I is determined from the luminaire I-table (see 5.2) after: corrections have been made for the orientation, tilt for calculation, and rotation of the luminaire as indicated in clause 6; and interpolation which follows the procedure in 5.3 has been applied; as well as any correction made that may be necessary because the light output of the lamp is temperature dependent and the luminaire is not used at the temperature at which it was photometrically measured.

Likewise, r for the appropriate value of *tan* ε and β is determined after the use of quadratic interpolation, if necessary.

7.1.2 Total luminance at a point

The total luminance at a point, L_P, is the sum of the contributions, L_1, L_2, L_3…….. L_n, from all the luminaires.

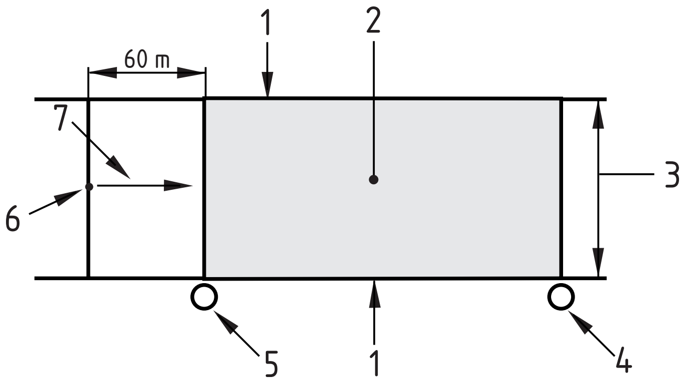
$$L_P = L_1 + L_2 + ... + L_k + ... + L_n = \sum_{k=1}^{n} L_k \tag{32}$$

7.1.3 Field of calculation for luminance

In the longitudinal direction of the relevant area, the field of calculation shall enclose two luminaires in the same row (Figure 9), the first luminaire being located 60 m ahead of the observer.

When there is more than one row of luminaires and the spacing of the luminaires differs between rows, the field of calculation shall lie between two luminaires in the row with the larger or largest spacing.

NOTE This procedure may not give accurate luminances for the whole installation as luminances will differ in the different spans between adjacent luminaires. It can be preferable to calculate luminances and uniformities over a longer longitudinal distance and consider a number of observer positions.



Key

1 Edge of relevant area

2 Field of calculation

3 Width of relevant area W_r

4 Last luminaire in field of calculation

5 First luminaire in field of calculation

6 Observer

7 Observation direction

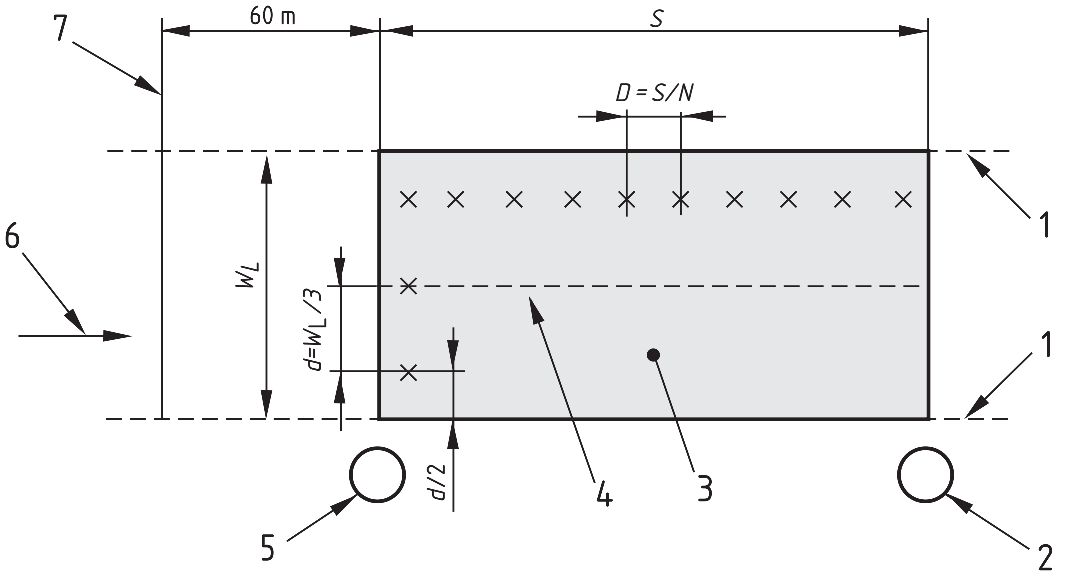
Figure 9 — Information for luminance calculations; field of luminance calculations for the relevant area

7.1.4 Position of calculation points

The calculation points shall be evenly spaced in the field of calculation as shown in Figure 10.

The first and last transverse rows of calculation points are spaced at one half the longitudinal spacing between points from the boundaries of the calculation field (Figure 10).

NOTE This grid is similar to the grid used for illuminance calculations as regards the positioning of the first and last row of calculation points in the transverse direction (Figure 15).



Key

1 Edge of lane

2 Last luminaire in calculation field

3 Field of calculation

4 Centre-line of lane

5 First luminaire in calculation field

6 Observation direction

7 Observer's longitudinal position

X denotes lines of calculation points in the transverse and longitudinal directions.

Figure 10 — Information for luminance calculations; position of calculation points in a driving lane

The spacing of the points in the longitudinal and transverse directions shall be determined as follows:

a) In the longitudinal direction

$$D = \frac{S}{N} \tag{33}$$

where:

D is the spacing between points in the longitudinal direction, in metres

S is the spacing between luminaires in the same row, in metres

N is the number of calculation points in the longitudinal direction with the following values:

for $S \leq 30$ m, $N = 10$;

for S > 30 m, the smallest integer giving $D \leq 3$ m. The first transverse row of calculation points is spaced at a distance D/2 beyond the first luminaire (remote from the observer).

b) In the transverse direction

The spacing (d) in the transverse direction is determined from the equation:

$$d = \frac{W_L}{3} \tag{34}$$

where:

d is the spacing between points in the tranverse direction, in metres

W_L is the width of the lane, in metres

The outermost calculation points are spaced d/2 from the edges of the lane.

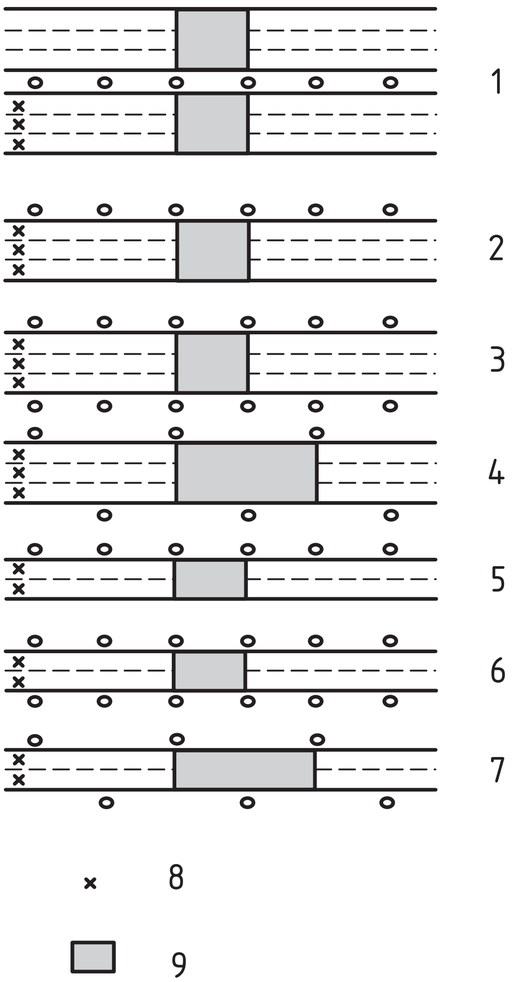
Where there is a hard shoulder and luminance information is required, the number and spacing of the calculation points shall be the same as for a driving lane.

7.1.5 Position of observer

For luminance calculations the observer´s eye is 1,5 m above the road level.

In the transverse direction the observer shall be positioned in the centre of each lane in turn. Average luminance (see 8.2), overall uniformity of luminance (see 8.3) and threshold increment (see 8.5) shall be calculated for the entire carriageway for each position of the observer. Longitudinal uniformity of luminance (see 8.4) shall be calculated for each centre-line. The operative values of average luminance, overall uniformity of luminance, and longitudinal uniformity of luminance shall be the lowest in each case; the operative value of threshold increment shall be the highest value.

Figure 11 gives examples of the observer position in relation to the field of calculation.



Key

1 Six lane road with central reservation

2 Three lane road. Single side luminaire arrangement

3 Three lane road. Double side luminaire arrangement

4 Three lane road. Staggered luminaire arrangement

5 Two lane road. Single side luminaire arrangement

6 Two lane road. Double side luminaire arrangement

7 Two lane road. Staggered luminaire arrangement

8 Observer position

9 Calculation field

Figure 11 — Examples of positions of observation points in relation to the field of calculation

7.1.6 Luminaires included in calculation

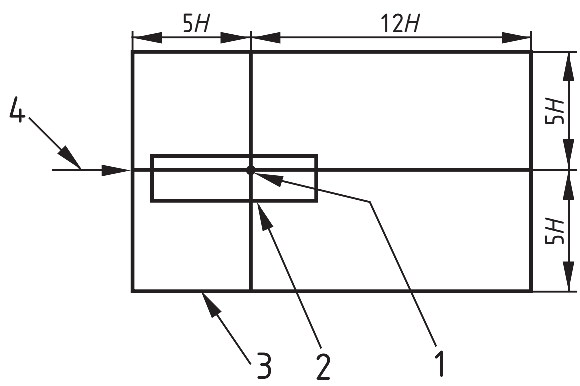
The boundary of the area for locating luminaires to be included in calculating the luminance at a point is determined as follows (Figure 12):

a) boundary on either side of the observer: at least five times the mounting height *H* on either side of the calculation point;

b) boundary furthest from the observer: at least 12*H* from the calculation point in the direction remote from the observer;

c) boundary nearest to the observer: at least 5*H* from the calculation point in the direction towards the observer.

NOTE The extent of these boundaries is governed by the area covered on the road by the *r*-table.



Key

1 Calculation point

2 Boundary of field of calculation

3 Boundary of area for location of luminaires

4 Observation direction

Figure 12 — Boundary of area in which luminaires are located for calculating the luminance at a point

7.2 Illuminance

7.2.1 General

In this standard any of four measures of illuminance may need to be calculated, depending on the design criteria chosen from EN 13201-2. These may be:

— horizontal illuminance;

— hemispherical illuminance;

— semicylindrical illuminance;

— vertical illuminance.

7.2.2 Horizontal illuminance at a point

Calculation points shall be located in a plane at ground level in the relevant area.

The horizontal illuminance at a point shall be calculated from the formula or a mathematically equivalent formula:

(35)

where:

E is the maintained horizontal illuminance at the point, in lux

I is the intensity in the direction of the point, in candelas per kilolumen

ε is the angle of incidence of the light at the point, in degrees

H is the mounting height of the luminaire, in metres

Φ is the initial luminous flux of the lamp or lamps in the luminaire, in kilolumens

MF is the product of the lamp flux maintenance factor and the luminaire maintenance factor

7.2.3 Hemispherical illuminance at a point

Calculation points shall be located in a plane at ground level in the relevant area.

The hemispherical illuminance at a point shall be calculated from the formula or a mathematically equivalent formula:

$$E = \frac{I \times [\cos^3 \varepsilon + \cos^2 \varepsilon] \times \Phi \times MF}{4 \times H^2} \qquad (36)$$

where:

E is the maintained hemispherical illuminance at the point, in lux

I is the intensity in the direction of the point, in candelas per kilolumen

ε is the angle of incidence of the light at the point

H is the mounting height of the luminaire, in metres

Φ is the initial luminous flux of the lamp or lamps in the luminaire, in kilolumens

MF is the product of the lamp flux maintenance factor and the luminaire maintenance factor

7.2.4 Semicylindrical illuminance at a point

Calculation points shall be located in a plane 1,5 m above the surface in the relevant area.

Semicylindrical illuminance varies with the direction of interest. The vertical plane in Figure 13, at right angles to the rear flat surface, shall be oriented parallel to the main directions of pedestrian movement, which for a road are usually longitudinal.

The semicylindrical illuminance at a point shall be calculated from the formula or a mathematically equivalent formula:

$$E = \frac{I \times [1 + \cos\alpha] \times \cos^2\varepsilon \times \sin\varepsilon \times \Phi \times MF}{\pi \times (H - 1{,}5)^2} \qquad (37)$$

where:

E is the maintained semicylindrical illuminance at the point, in lux

I is the intensity in the direction of the point, in candelas per kilolumen

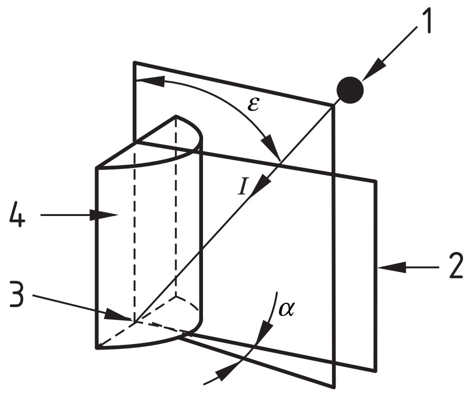
α is the angle between the vertical plane containing the incident light path and the vertical plane at right-angles to the flat surface of the semicylinder, as shown in Figure 13

ε is the angle of incidence of the light to the normal to the horizontal plane, at the point

H is the mounting height of the luminaire, in metres

Φ is the initial luminous flux of the lamp or lamps in the luminaire, in kilolumens

MF is the product of the lamp flux maintenance factor and the luminaire maintenance factor



Key

1 Luminaire

2 Vertical plane at right-angles to flat surface of semicylinder

3 Calculation point

4 Flat surface of semicylinder

Figure 13 — Angles used in the calculation of semicylindrical illuminance

7.2.5 Vertical iluminance at a point

Calculation points shall be located in a plane 1,5 m above the surface in the relevant area.

Vertical illuminance varies with the direction of interest. The vertical illumination plane in Figure 14 shall be oriented at right-angles to the main directions of pedestrian movement, which for a road are usually up and down the road.

The vertical illuminance at a point shall be calculated from the formula or a mathematically equivalent formula:

$$E = \frac{I \times \cos^2 \varepsilon \times \sin \varepsilon \times \cos \alpha \times \Phi \times MF}{(H - 1{,}5)^2} \qquad (38)$$

where:

E is the maintained vertical illuminance at the point, in lux

I is the intensity in the direction of the point, in candelas per kilolumen

α is the angle in degrees between the vertical plane containing the incident light path and the vertical plane at right-angles to the vertical plane of calculation, as shown in Figure 14

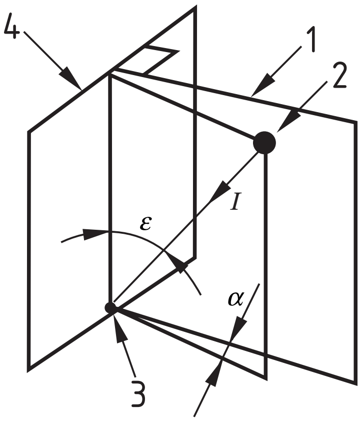
ε is the angle of incidence of the light to the horizontal plane, at the point, in degrees

H is the mounting height of the luminaire, in metres

ϕ is the initial luminous flux of the lamp or lamps in the luminaire, in kilolumens

MF is the product of the lamp flux maintenance factor and the luminaire maintenance factor

This formula is valid only for $\varepsilon \leq 90°$ and $\alpha \leq 90°$.



Key

1 Vertical plane at right-angles to vertical illumination plane

2 Luminaire

3 Calculation point

4 Vertical illumination plane

Figure 14 — Angles used in the calculation of vertical illuminance

7.2.6 Total illuminance at a point

The total illuminance at a point, E_P, is the sum of the contributions, E_1, E_2, E_3,……… E_n, from all the luminaires.

$$E_P = E_1 + E_2 + ... + E_k + ... + E_n = \sum_{k=1}^{n} E_k \qquad (39)$$

NOTE Only illuminance measures of the same type can be summed. Moreover, these should have the same directionality.

7.2.7 Field of calculation for illuminance

The field of calculation shall be the same as that indicated in Figure 11.

NOTE To economize on computer processing time, for staggered installations the calculation field can be taken between consecutive luminaires on opposite sides of the road without affecting the result.

7.2.8 Position of calculation points

The calculation points shall be evenly spaced in the field of calculation (Figure 15) and their number shall be chosen as follows:

a) In the longitudinal direction

The spacing in the longitudinal direction shall be determined from the equation:

$$D = \frac{S}{N} \tag{40}$$

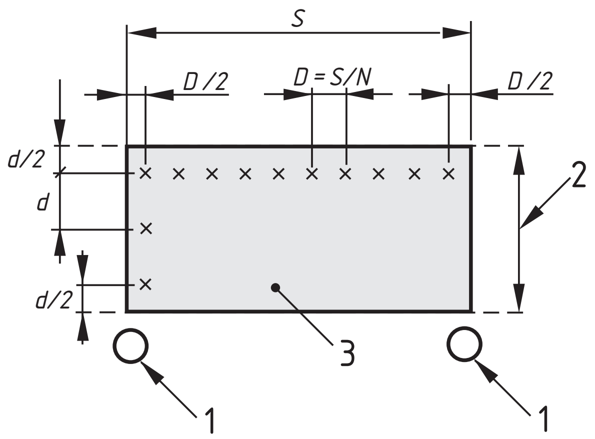
where:

D is the spacing between points in the longitudinal direction, in metres;

S is the spacing between luminaires, in metres;

N is the number of calculation points in the longitudinal direction with the following values:
for $S \leq 30$ m, N = 10;
for $S > 30$ m, the smallest integer giving $D \leq 3$ m.

The first row of calculation points is spaced at a distance $D/2$ (in metres) beyond the first luminaire.



Key

1 Luminaire

2 Width of relevant area W_r

3 Field of calculation

x denotes lines of calculation points in the transverse and longitudinal directions

Figure 15 — Information for illuminance calculations; calculation points on relevant area

b) In the transverse direction

$$d = \frac{W_r}{n} \tag{41}$$

where:

- *d* is the spacing between points in the transverse direction, in metres;
- W_r is the width of the carriageway or relevant area, in metres;
- *n* is the number of points in the transverse direction with a value greater or equal to 3 and is the smallest integer giving $d \leq 1{,}5$ m.

The spacing of points from the edges of the relevant area is $D/2$ in the longitudinal direction, and $d/2$ in the transverse direction, as indicated in Figure 15.

7.2.9 Luminaires included in calculation

Luminaires that are situated within five times the mounting height from the calculation point shall be included in the calculation.

7.2.10 Illuminance on areas of irregular shape

For these areas it may be necessary to choose a rectangular calculation field which encloses and is therefore larger than the relevant area. Grid points used for the calculation of the quality characteristics should be chosen from those points which lie within the boundary of the relevant area. When the spacing of the luminaires is not regular it may not be possible to link the spacing of the grid points to the spacing of the luminaires, but the spacing in either direction shall not exceed 1,5 m. The principal directions of traffic flow for the calculation of vertical illuminance and semicylindrical illuminance should be decided after considering the use or likely use of the area.

8 Calculation of quality characteristics

8.1 General

Quality characteristics relating to luminance or illuminance shall be obtained from the calculated grids of luminance or illuminance without further interpolation. If the grid points do not coincide with the centre of lanes, for the calculation of longitudinal uniformity of luminance it will be necessary to calculate the luminance of points on the centreline of each lane and the hard shoulder, if present, in accordance with 8.4.

For initial average illuminance or initial average luminance, *MF* is 1,0 and initial values of the luminous flux of the lamp or lamps in the luminaires shall be taken. For average luminance or average illuminance after a stated period, the *MF* for the luminaire after the stated period in the environmental conditions of the installation shall be taken together with the luminous flux in kilolumens of the light source or sources in the luminaire after the stated period.

8.2 Average luminance

The average luminance shall be calculated as the arithmetic mean of the luminances at the grid points in the field of calculation.

8.3 Overall uniformity

The overall uniformity shall be calculated as the ratio of the lowest luminance, occurring at any grid point in the field of calculation, to the average luminance.

8.4 Longitudinal uniformity

The longitudinal uniformity shall be calculated as the ratio of the lowest to the highest luminance in the longitudinal direction along the centre line of each lane, and the hard shoulder in the case of motorways (Figure 11). The number of points in the longitudinal direction *(N)* and the spacing between them shall be the same as those used for the calculation of average luminance.

The observer's position shall be in line with the row of calculation points.

8.5 Threshold increment

The threshold increment *(TI)* is calculated from the equations or mathematically equivalent equations:

$$TI = \frac{65}{(\textit{average road luminance})^{0,8}} \times L_{\mathrm{v}} \ \% \tag{42}$$

$$L_{\mathrm{v}} = 10 \sum_{\mathrm{k}=1}^{n} \frac{E_{\mathrm{k}}}{\theta_{\mathrm{k}}^2} = \frac{E_1}{\theta_1^2} + \frac{E_2}{\theta_2^2} + \ldots.. + \frac{E_{\mathrm{k}}}{\theta_{\mathrm{k}}^2} + \ldots.. + \frac{E_{\mathrm{n}}}{\theta_{\mathrm{n}}^2} \tag{43}$$

where:

the initial *average road luminance* (in cd/m^2) is the average road luminance calculated for luminaires in their new state and for lamps emitting the initial lamp flux, in lumens;

L_{v} is the equivalent veiling luminance, in candelas per square metre;

E_{k} is the illuminance (in lux, based on the initial lamp flux, in lumens) produced by the *k*th luminaire in its new state on a plane normal to the line of sight and at the height of the observer's eye;

The observer's eye, height 1,5 m above road level, is positioned in the centre line of each lane in turn, as indicated in Figure 11, and longitudinally at a distance in metres of 2,75 (H - 1,5), where H is the mounting height (in metres), in front of the field of calculation. The line of sight is 1° below the horizontal and in a vertical plane in the longitudinal direction passing through the observer's eye.

θ_{k} is the angle, in degrees, of arc between the line of sight and the line from the observer to the centre of the kth luminaire.

The summation is performed for the first luminaire in the direction of observation and luminaires beyond, up to a distance of 500 m in each luminaire row, and stopped when a luminaire in that row gives a contribution to the veiling luminance which is less than 2 % of the total veiling luminance of the preceding luminaires in the row. Luminaires above a screening plane which is inclined at 20° to the horizontal, and which passes through the observer's eye, and which intersects the road in a transverse direction, shall be excluded from the calculation.

The calculation is commenced with the observer in the initial position stated above, and repeated with the observer moved forward in increments that are the same in number and distance as are used for the longitudinal spacing of luminance points. The procedure is repeated with the observer positioned in the centre line of each lane using in each case the *initial average road luminance* appropriate to the observer position.

The maximum value of *TI* found is the operative value.

This equation is valid for 0,05 < *average road luminance* < 5 cd/m^2 and 1,5 < θ_{k} <60 degrees of arc.

NOTE The constant 10 in equation 43 is valid for a 23 year old observer. Constants for other ages can be calculated from the formula:

$$9{,}86 \cdot \left[1 + \left(\frac{A}{66{,}4}\right)^4\right]$$

where A is the age of the observer, in years.

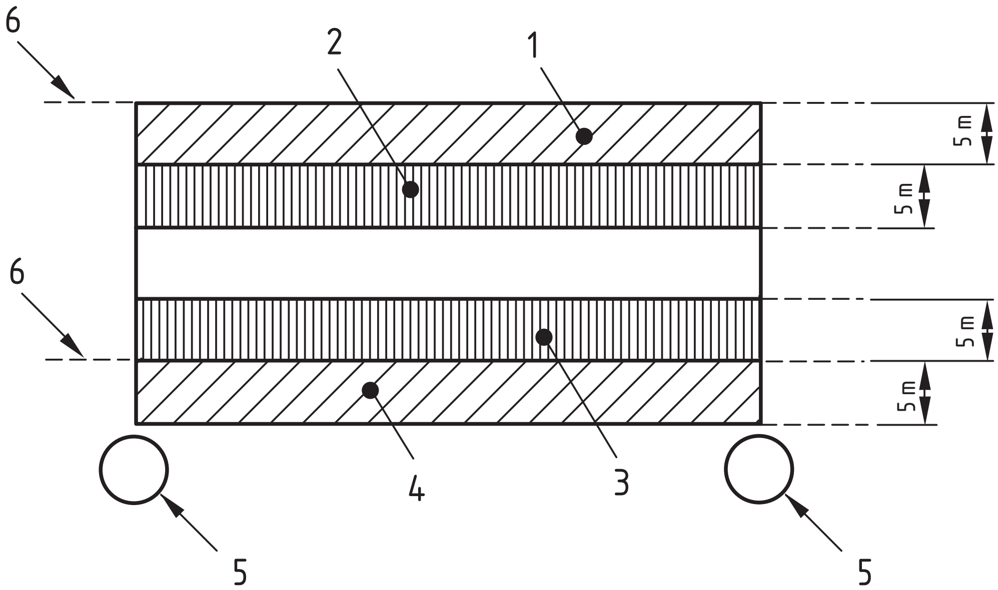
8.6 Surround ratio

The surround ratio is the average horizontal illuminance on the two longitudinal strips each adjacent to the two edges of the carriageway, and lying off the carriageway, divided by the average horizontal illuminance on two longitudinal strips each adjacent to the two edges of the carriageway, but lying on the carriageway. The width of all four strips shall be the same, and equal to 5 m, or half the width of the carriageway, or the width of the unobstructed strip lying off the carriageway, whichever is the least. For dual carriageways, both carriageways together are treated as a single carriageway unless they are separated by more than 10 m.

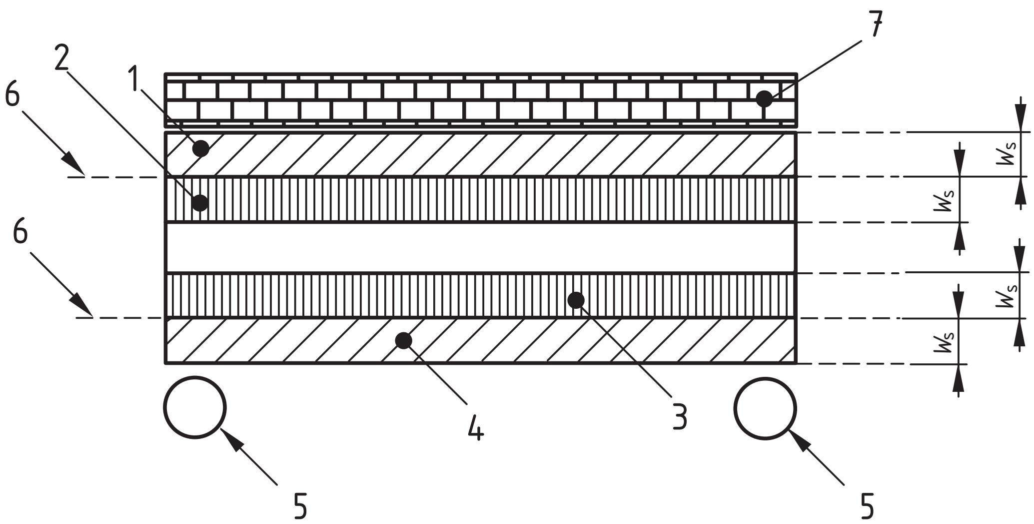
The horizontal illuminance shall be calculated by the procedure specified in 7.2.2. The field of calculation shall be as indicated in 7.2.7. The number of luminaires considered shall be the same as indicated in 7.2.9. The position of the calculation points within each strip shall be as indicated in 7.2.8.

Figure 16 gives examples of the location of the strips and their location for the calculation of surround ratio. For this figure, the following equation applies:

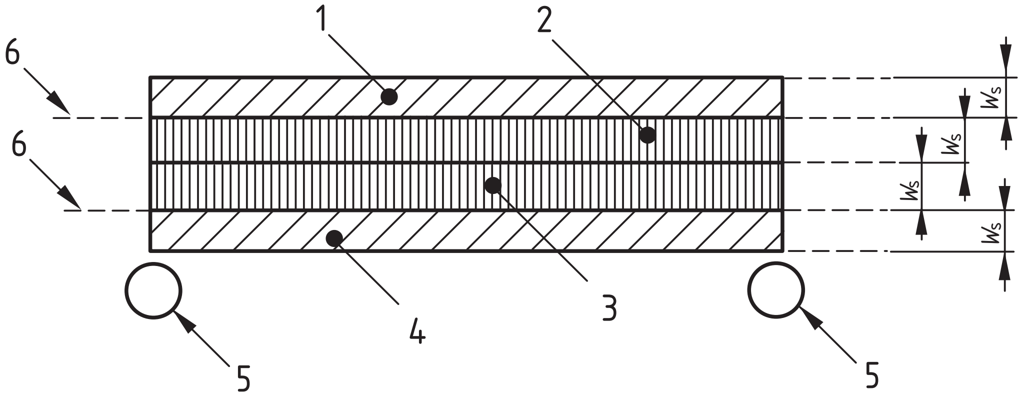
$$Surround\ ratio = \frac{Average\ illuminance\ of\ strip\ 1 + Average\ illuminance\ of\ strip\ 4}{Average\ illuminance\ of\ strip\ 2 + Average\ illuminance\ of\ strip\ 3}$$



a) Width of strip equals 5 m



b) Width of strip less than 5 m because of obstruction



c) Width of strip less than 5 m because width of carriageway is less than 10 m

Key

1 Strip 1

2 Strip 2

3 Strip 3

4 Strip 4

5 Luminaire

6 Edge of carriageway

7 Obstruction

W_S Width of strip

Figure 16 — Location and width of strips for calculating surround ratio

8.7 Measures of illuminance

8.7.1 General

Measures of illuminance include horizontal plane, vertical plane, hemispherical, and semicylindrical illuminance.

8.7.2 Average illuminance

The average illuminance shall be calculated as the arithmetic mean of the illuminances at the grid points in the field of calculation.

For conflict, pedestrian, and other irregularly shaped areas, the procedure in 7.2.10 shall be followed.

8.7.3 Minimum illuminance

The minimum illuminance shall be taken as the lowest illuminance occurring at any grid point in the field of calculation of illuminance.

8.7.4 Uniformity of illuminance

The uniformity of illuminance shall be calculated as the ratio of the lowest illuminance, occurring at any grid point in the field of calculation, to the average illuminance.

9 Ancillary data

When photometric performance data are prepared for an installation, the following ancillary data shall be declared:

a) identification of the luminaires;

b) identification of *I*-table;

c) identification of the *r*-table with a clear declaration of the value of Q_0 used; not required when the calculations are solely those of illuminance;

d) tilt during measurement of the luminaires;

e) tilt in application of the luminaires;

f) rotation of the luminaires, if different from zero;

g) orientation of the luminaires, if different from zero;

h) identification of the light sources;

i) luminous flux of the light sources on which the calculations are based;

j) maintenance factors applied;

k) definition of the area of calculation;

l) position of the luminaires in plan or a numerical description;

m) mounting height of the luminaires;

n) direction of interest for vertical illuminance and semicylindrical illuminance;

o) any deviations from the procedures given in this standard, including the calculation of threshold increment for an observer of other than 23 years old.

Bibliography

prCEN/TR 13201-1*, Road lighting —Selection of lighting classes.*

EN 12665, *Light and lighting — Basic terms and criteria for specifying lighting requirements.*

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